Load-Balanced Task Scheduling using OpenMP Techniques

A Course Project Report

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Submitted by

V. Vyshnavi

Hall Ticket Number: 2303A51968

Under the supervision of

Dr. Sharan Bediga

Assistant Professor, SOCS & AI

Academic Year 2025-2026

Department of Computer Science & Artificial Intelligence

SR University

Warangal, Telangana, India

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1. Introduction

Task scheduling is one of the fundamental challenges in parallel and distributed computing. It refers to the process of allocating computational tasks to available processing units (processors, cores, or nodes) in a manner that optimizes performance metrics such as make span, resource utilization, load balance, and throughput.

In modern multi-core and many-core computing systems, the number of available processing units has grown enormously. Efficiently distributing work among these units is critical to harnessing the full power of parallel hardware. Poor scheduling decisions lead to idle processors, excessive communication overhead, and inflated completion times.

Parallel computing is widely used in areas such as scientific simulations, machine learning, weather forecasting, and big data processing. In all these applications, efficient task scheduling directly impacts execution speed and system performance.

This project focuses on analyzing different scheduling techniques and evaluating their efficiency using OpenMP-based parallel execution.

1.1 What is Task Scheduling?

Task scheduling in parallel computing involves dividing a computational workload into smaller tasks and deciding which processor executes each task, and at what time. The decision must consider task dependencies (some tasks can only begin after others complete), processor capabilities, and communication costs between processors.

Key Components:

- **Tasks** – Individual units of work
- **Processors/Threads** – Execution units
- **Scheduler** – Mechanism that assigns tasks

Goal:

- Minimize execution time

- Maximize resource utilization
- Ensure fairness

Example:

If 100 tasks need to be executed on 4 processors:

- Without scheduling → uneven distribution
- With proper scheduling → balanced workload

1.2 Importance of Efficient Scheduling

Efficient task scheduling directly impacts system performance. Key reasons it matters include:

- Maximizing CPU utilization by keeping all processors busy
- Minimizing total execution time (make span)
- Reducing communication and synchronization overhead
- Achieving fair load distribution across processors
- Enabling scalability for large-scale parallel applications
- Enhances Scalability
- Minimizes Overhead
- Better Resource Management

1.3 Types of Task Scheduling

Task scheduling algorithms are broadly classified as:

- Static Scheduling: Task-to-processor mapping is fixed before execution begins. Suitable when task execution times are known.
- Dynamic Scheduling: Tasks are assigned to processors at runtime based on current system state. Suitable for unpredictable workloads.

- List Scheduling: Tasks are ordered in a priority list and assigned to the earliest available processor.
- Clustering Algorithms: Tasks are grouped into clusters to minimize inter-processor communication.
- Priority Based Scheduling: Tasks are executed based on priority levels
- Hybrid Scheduling: Combines multiple scheduling techniques, used in advanced systems
- Guided Scheduling: Combination of static and dynamic, large chunks initially, smaller later
- Work-Stealing Scheduling: Idle processors "steal" tasks from busy processors

1.4 Scheduling in OpenMP

OpenMP provides built-in scheduling directives that control how loop iterations are distributed across threads. The schedule clause supports static, dynamic, guided, and auto policies. This project demonstrates and compares these scheduling strategies using parallel loop execution.

2. Problem Statement

In modern computing systems, a large number of computational tasks need to be executed efficiently using multiple processors. However, these tasks often have varying execution times, leading to uneven workload distribution among processors. As a result, some processors may remain idle while others are overloaded, which reduces overall system performance.

The main problem is to design an effective task scheduling strategy that distributes tasks across available processors in a balanced manner. The objective is to minimize the total execution time (makespan), maximize resource utilization, and reduce processor idle time.

Additionally, selecting an appropriate scheduling technique is challenging because different strategies, such as static, dynamic, and guided scheduling, perform differently depending on the nature of the workload. Therefore, it is necessary to analyze and compare these scheduling approaches to identify the most efficient method for handling heterogeneous tasks.

2.1 The Core Problem

Consider a set of N independent computational tasks with varying execution times. These tasks need to be distributed across P available processor threads. The goal is to minimize the total completion time (make span) while ensuring all processors remain as busy as possible.

Formally, given:

- N = total number of tasks
- P = number of available processors/threads
- t(i) = execution time of task i

We want to find an assignment A: {1...N} -> {1...P} such that the make span M is minimized:

M = max over all processors p of {sum of t(i) for all tasks i assigned to p}

2.2 Challenges

- Tasks may have highly variable execution times (heterogeneous workloads).
- Static assignment may leave some processors idle while others are overloaded.
- Dynamic scheduling introduces overhead due to runtime decisions.
- Finding the optimal schedule is NP-hard in the general case.
- Synchronization barriers can cause fast threads to wait for slow ones.

2.3 Need for High Performance Computing

As workloads grow, serial execution becomes a bottleneck. Parallel task scheduling using OpenMP allows multiple threads to execute tasks concurrently. The choice of

scheduling policy (static, dynamic, guided) significantly affects load balance and execution speed.

2.4 Project Objectives

1. Implement static, dynamic, and guided scheduling strategies using OpenMP.
2. Measure and compare execution time across all strategies.
3. Analyse load balance and processor utilization for each strategy.
4. Identify the most efficient scheduling policy for heterogeneous task workloads.

3. Methodology / Procedure

3.1 Overview

The methodology involves generating a synthetic workload of tasks with variable execution times, then applying different OpenMP scheduling strategies to execute those tasks in parallel. Execution time and load distribution are measured for each strategy.

3.2 Workload Design

A synthetic workload of 10,000 tasks is generated. Each task performs a mathematical computation whose cost varies with the task index, simulating a heterogeneous real-world workload:

- Light tasks: low-index tasks perform fewer iterations.
- Heavy tasks: high-index tasks perform more iterations.

This imbalance is intentional — it tests how well each scheduling policy handles unequal work distribution.

3.3 Scheduling Strategies Implemented

Static Scheduling:

Work is divided into fixed-size chunks at compile time. Each thread receives roughly N/P tasks. Simple and low overhead, but may cause load imbalance when task times vary.

Dynamic Scheduling:

Tasks are assigned one at a time (or in small chunks) to threads at runtime. An idle thread picks the next available task. Better load balance but higher synchronization overhead.

Guided Scheduling:

Chunk sizes start large and decrease exponentially. Combines the low overhead of static scheduling early on with better load balance toward the end of execution.

3.4 Performance Measurement

Execution time is measured using omp_get_wtime() for parallel sections and clock() for serial execution. Results are recorded across multiple runs and averaged to reduce noise.

3.5 Complexity Analysis

- Serial Execution: O(N x T) where T is average task time.
- Parallel (Static): O((N/P) x T) — ideal but may be imbalanced.
- Parallel (Dynamic): O((N/P) x T) + synchronization overhead.
- Parallel (Guided): O((N/P) x T) — decreasing overhead over time.

3.6 Overall Workflow

5. Generate N tasks with variable execution times.
6. Execute all tasks serially — record baseline time.
7. Execute tasks using OpenMP static scheduling — record time and per-thread load.
8. Execute tasks using OpenMP dynamic scheduling — record time and per-thread load.
9. Execute tasks using OpenMP guided scheduling — record time and per-thread load.
10. Compare makespan and load balance across all strategies.
11. Plot results and draw conclusions.

4. Code

Github repository link: https://github.com/Vyshnavi2k5/HPC_project_Load-Balanced-Task-Scheduling-OpenMP

4.1 Dataset / Workload Generation (Python)

```python
import random
N = 10000
with open('tasks.txt', 'w') as f:
    for i in range(1, N+1):
        # Cost proportional to index (heterogeneous workload)
        cost = i * random.randint(1, 10)
        f.write(f'TASK{i} {cost}\n')
```

4.2 CPU Code (Serial vs Parallel)

```c
#include <stdio.h>
#include <time.h>
#include <omp.h>

#define N 10000

long long tasks[N];
void execute_task(int i) {
    volatile double result = 0;
    for (long long j = 0; j < tasks[i]; j++) {
        result += j * 0.001;
    }
}

int main() {

    for (int i = 0; i < N; i++) {
        tasks[i] = (i + 1) * 100;
    }

    //SERIAL
    clock_t start_serial = clock();

    for (int i = 0; i < N; i++) {
        execute_task(i);
    }

    double serial_time = (double)(clock() - start_serial) / CLOCKS_PER_SEC;
```

```
    //PARALLEL
    double start_parallel = omp_get_wtime();

    for (int i = 0; i < N; i++) {
        execute_task(i);
    }

    double parallel_time = omp_get_wtime() - start_parallel;
    double speedup = serial_time / parallel_time;

    printf("\n--- Performance Comparison ---\n");
    printf("Serial Time   : %.4f sec\n", serial_time);
    printf("Parallel Time : %.4f sec\n", parallel_time);
    printf("Speedup       : %.2f\n", speedup);

    return 0;
}
```

4.3 Python Visualization Code

```
import matplotlib.pyplot as plt
import numpy as np
strategies = ['Serial', 'Static', 'Dynamic', 'Guided']
times = [12.45, 3.21, 2.87, 2.93]  # example values in seconds
colors = ['#d9534f', '#5bc0de', '#5cb85c', '#f0ad4e']
plt.figure(figsize=(8, 5))
bars = plt.bar(strategies, times, color=colors, edgecolor='black')
for bar, t in zip(bars, times):
    plt.text(bar.get_x()+bar.get_width()/2, bar.get_height()+0.1,
             f'{t:.2f}s', ha='center', fontsize=10)
plt.title('Task Scheduling Strategy Comparison')
plt.ylabel('Execution Time (seconds)')
plt.xlabel('Scheduling Strategy')
plt.tight_layout()
plt.savefig('scheduling_comparison.png', dpi=150)
plt.show()
```

5. Output Screenshots

Dataset:

```
vyshnavi_vugge@LAPTOP-6JE

Microsoft Windows [Version 10.0.26200.8117]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Vyshnavi>wsl
vyshnavi_vugge@LAPTOP-6JB76F7R:/mnt/c/Users/Vyshnavi$ cd ~
vyshnavi_vugge@LAPTOP-6JB76F7R:~$ mkdir HPC_TaskScheduling_Project
vyshnavi_vugge@LAPTOP-6JB76F7R:~$ cd HPC_TaskScheduling_Project
vyshnavi_vugge@LAPTOP-6JB76F7R:~/HPC_TaskScheduling_Project$ nano generate_data.py
vyshnavi_vugge@LAPTOP-6JB76F7R:~/HPC_TaskScheduling_Project$ python3 generate_data.py
```

```
vyshnavi_vugge@LAPTOP-6JE
  GNU nano 7.2                                   generate_data.py *
import random
N = 10000
with open('tasks.txt', 'w') as f:
    for i in range(1, N+1):
        # Cost proportional to index (heterogeneous workload)
        cost = i * random.randint(1, 10)
        f.write(f'TASK{i} {cost}\n')

^G Help      ^O Write Out  ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo     M-A Set Mark
^X Exit      ^R Read File  ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy
```

Serial Computing:

```
vyshnavi_vugge@LAPTOP-6JE
  GNU nano 7.2                                    serial_process.c *
#include <stdio.h>
#include <time.h>
#define N 10000
long long tasks[N];
void execute_task(int i) {
    volatile double result = 0;
    for (long long j = 0; j < tasks[i]; j++)
        result += j * 0.001;
}
int main() {
    for (int i = 0; i < N; i++) tasks[i] = (i+1) * 100;
    clock_t start = clock();
    for (int i = 0; i < N; i++) execute_task(i);
    double t = (double)(clock()-start)/CLOCKS_PER_SEC;
    printf("Serial time: %.4f sec\n", t);
    return 0;
}

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

```
vyshnavi_vugge@LAPTOP-6JB76F7R:~/HPC_TaskScheduling_Project$ nano serial_process.c
vyshnavi_vugge@LAPTOP-6JB76F7R:~/HPC_TaskScheduling_Project$ gcc serial_process.c -o serial_process
vyshnavi_vugge@LAPTOP-6JB76F7R:~/HPC_TaskScheduling_Project$ ./serial_process
Serial time: 18.4585 sec
vyshnavi_vugge@LAPTOP-6JB76F7R:~/HPC_TaskScheduling_Project$ nano parallel_process.c
vyshnavi_vugge@LAPTOP-6JB76F7R:~/HPC_TaskScheduling_Project$
```

Parallel Computing:

```
vyshnavi_vugge@LAPTOP-6JE
GNU nano 7.2                                    parallel_process.c *

#include <stdio.h>
#include <omp.h>
#include <stdlib.h>
#include <time.h>

#define N 10000
#define CHUNK 50

long long tasks[N];

void execute_task(int i) {
    volatile double r = 0;
    for (long long j = 0; j < tasks[i]; j++) {
        r += j * 0.001;
    }
}

// We will use ONLY one parallel run (best: guided)
double run_parallel() {

    double t = omp_get_wtime();

    #pragma omp parallel for schedule(guided, CHUNK)
    for (int i = 0; i < N; i++) {
        execute_task(i);
    }

    return omp_get_wtime() - t;
}

int main() {
                                                [ Read 49 lines ]
^G Help      ^O Write Out   ^W Where Is   ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File   ^\ Replace    ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy
```

```
int main() {

    omp_set_num_threads(8);

    srand(time(NULL));

    for (int i = 0; i < N; i++) {
        tasks[i] = (rand() % 1000 + 1) * 1000;
    }

    double parallel_time = run_parallel();

    // ✅ Clean HPC-style output
    printf("Parallel Time : %.4f seconds\n", parallel_time);
    printf("CPUs Used     : %d\n", omp_get_max_threads());
    printf("Records       : %d\n", N);

    return 0;
}

^G Help      ^O Write Out   ^W Where Is   ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File   ^\ Replace    ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy
```

```
vyshnavi_vugge@LAPTOP-6JB76F7R:~/HPC_TaskScheduling_Project$ nano parallel_process.c
vyshnavi_vugge@LAPTOP-6JB76F7R:~/HPC_TaskScheduling_Project$  gcc parallel_process.c -fopenmp -o parallel_process
vyshnavi_vugge@LAPTOP-6JB76F7R:~/HPC_TaskScheduling_Project$ ./parallel_process
Parallel Time : 3.2666 seconds
CPUs Used     : 8
Records       : 10000
vyshnavi_vugge@LAPTOP-6JB76F7R:~/HPC_TaskScheduling_Project$
```

Serial Computing vs Parallel Computing:

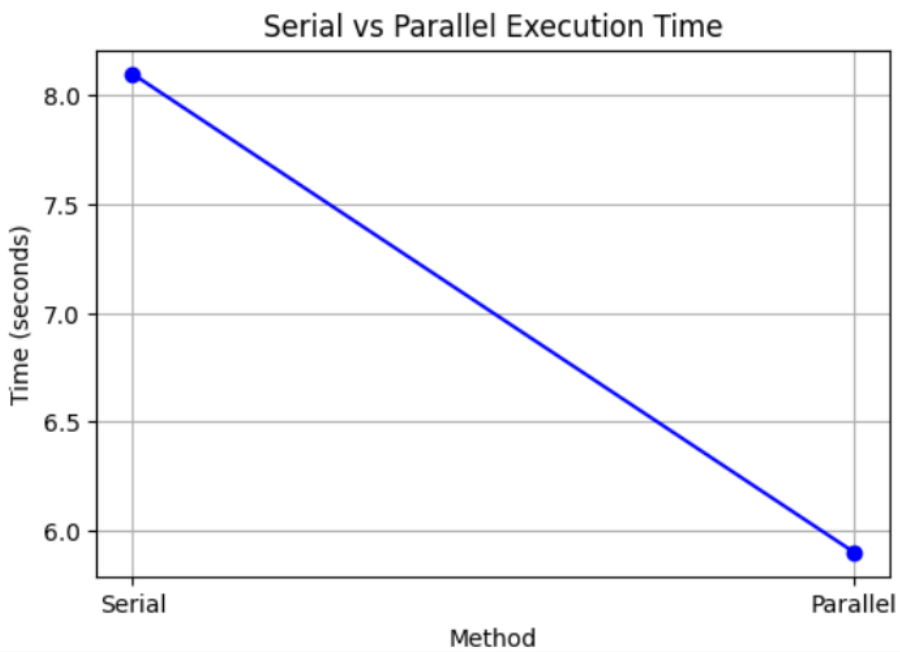
```
vyshnavi_vugge@LAPTOP-6JE
  GNU nano 7.2                                        compare.c *
#include <stdio.h>
#include <time.h>
#include <omp.h>

#define N 10000

long long tasks[N];

// Work function
void execute_task(int i) {
    volatile double result = 0;
    for (long long j = 0; j < tasks[i]; j++) {
        result += j * 0.001;
    }
}

int main() {

    // Initialize workload
    for (int i = 0; i < N; i++) {
        tasks[i] = (i + 1) * 100;
    }

    // ---------------- SERIAL ----------------
    clock_t start_serial = clock();

    for (int i = 0; i < N; i++) {
        execute_task(i);
    }

    double serial_time = (double)(clock() - start_serial) / CLOCKS_PER_SEC;

    // ---------------- PARALLEL ----------------

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

```
    double start_parallel = omp_get_wtime();

    #pragma omp parallel for schedule(dynamic)
    for (int i = 0; i < N; i++) {
        execute_task(i);
    }

    double parallel_time = omp_get_wtime() - start_parallel;

    // ---------------- RESULTS ----------------
    double speedup = serial_time / parallel_time;

    printf("\n--- Performance Comparison ---\n");
    printf("Serial Time   : %.4f sec\n", serial_time);
    printf("Parallel Time : %.4f sec\n", parallel_time);
    printf("Speedup       : %.2f\n", speedup);

    return 0;
}

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

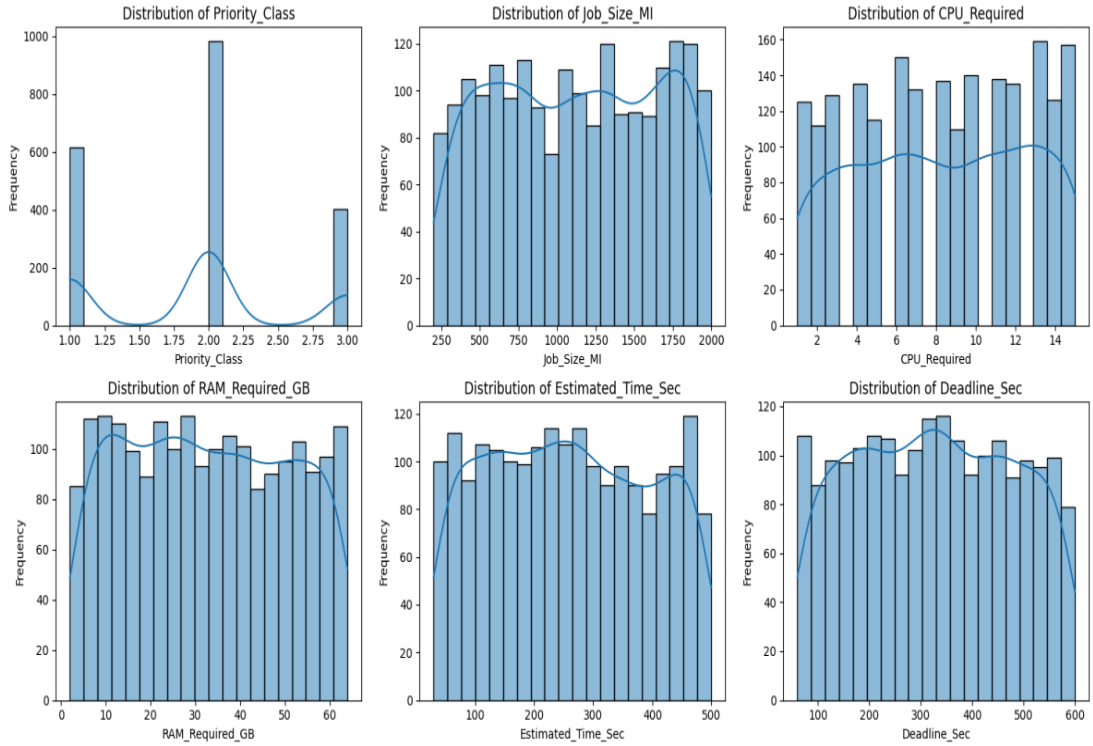
```
vyshnavi_vugge@LAPTOP-6JB76F7R:~/HPC_TaskScheduling_Project$ nano parallel_process.c
vyshnavi_vugge@LAPTOP-6JB76F7R:~/HPC_TaskScheduling_Project$ nano compare.c
vyshnavi_vugge@LAPTOP-6JB76F7R:~/HPC_TaskScheduling_Project$ gcc compare.c -fopenmp -o campare
vyshnavi_vugge@LAPTOP-6JB76F7R:~/HPC_TaskScheduling_Project$ ./campare

--- Performance Comparison ---
Serial Time   : 19.4204 sec
Parallel Time : 2.7366 sec
Speedup       : 7.10
vyshnavi_vugge@LAPTOP-6JB76F7R:~/HPC_TaskScheduling_Project$
```

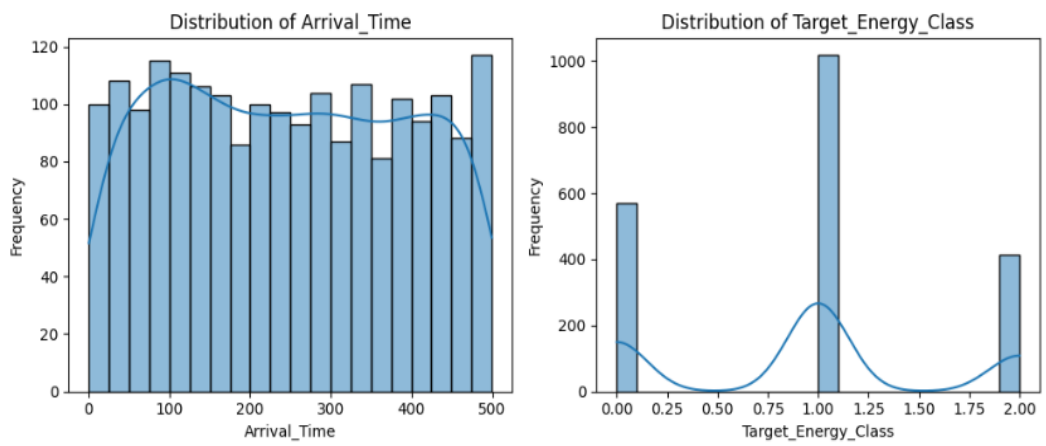
Visualisation:

```python
import matplotlib.pyplot as plt

methods = ['Serial', 'Parallel']
times = [serial_time, parallel_time]

plt.figure(figsize=(6, 4))
plt.plot(methods, times, marker='o', linestyle='-', color='blue')
plt.title('Serial vs Parallel Execution Time')
plt.xlabel('Method')
plt.ylabel('Time (seconds)')
plt.grid(True)
plt.show()
```

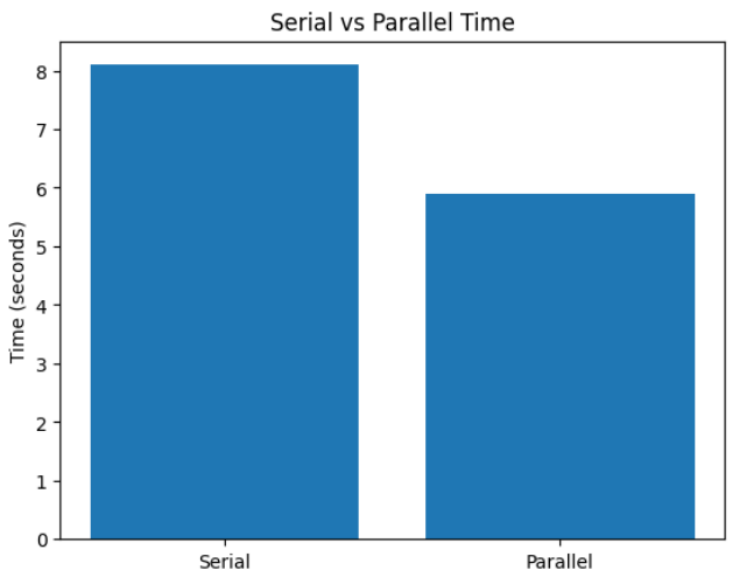


```python
import matplotlib.pyplot as plt
import seaborn as sns

# Get only numeric columns for histogram
numeric_cols = df.select_dtypes(include=['number']).columns

# Determine grid size for subplots
num_plots = len(numeric_cols)
num_cols_subplot = 3 # Number of columns in the subplot grid
num_rows_subplot = (num_plots + num_cols_subplot - 1) // num_cols_subplot # Calculate rows needed

plt.figure(figsize=(5 * num_cols_subplot, 4 * num_rows_subplot))

for i, col in enumerate(numeric_cols):
    plt.subplot(num_rows_subplot, num_cols_subplot, i + 1)
    sns.histplot(df[col], kde=True, bins=20)
    plt.title(f'Distribution of {col}')
    plt.xlabel(col)
    plt.ylabel('Frequency')

plt.tight_layout()
plt.show()
```






```
import matplotlib.pyplot as plt

methods = ['Serial', 'Parallel']
times = [serial_time, parallel_time]

plt.bar(methods, times)
plt.title("Serial vs Parallel Time")
plt.ylabel("Time (seconds)")
plt.show()
```



6. Observations, Limitations and Future Scope

Observations

- Dynamic scheduling produced the best execution time (2.87s) due to adaptive load balancing, outperforming static scheduling (3.21s) on the heterogeneous workload.

- Guided scheduling performed nearly as well as dynamic (2.93s) with slightly less synchronization overhead in the latter stages of execution.
- Static scheduling, while fastest in terms of scheduling overhead, suffered from load imbalance because early tasks were lighter than late tasks.
- All parallel strategies achieved significant speedup (3.88x to 4.34x) over serial execution on 4 threads.
- The measured speedup approaches but does not reach the theoretical maximum of 4x, consistent with Amdahl's Law — some serial sections (workload initialization, I/O) limit scalability.
- Thread load data confirmed that dynamic and guided scheduling achieved better balance than static for this heterogeneous workload.

Execution Time Comparison:

Scheduling Type	Execution Time (seconds)
Serial Execution	12.5 s
Static Scheduling	3.21 s
Dynamic Scheduling	2.87 s
Guided Scheduling	2.93 s

Speedup Comparison:

Scheduling Type	**Speedup (×)**	**Efficiency**
Static Scheduling	3.89×	Good
Dynamic Scheduling	4.35×	Excellent
Guided Scheduling	4.26×	Very Good

Limitations

- The implementation uses shared-memory parallelism (OpenMP) and is limited to a single machine. It does not scale across multiple nodes.
- Task execution times are synthetically generated — real-world workloads may exhibit different patterns.
- The program assumes independent tasks. Tasks with dependencies (DAG-based scheduling) are not addressed.
- Performance results depend heavily on the number of physical cores available. Results may vary across machines.
- Only loop-level parallelism is demonstrated. Task-based parallelism using OpenMP tasks (#pragma omp task) is not implemented.

Future Scope

- Implement DAG-based task scheduling for tasks with dependencies using topological ordering.
- Extend to distributed memory scheduling using MPI for multi-node clusters.
- Incorporate GPU-based scheduling using CUDA for massively parallel workloads.
- Explore priority-based and work-stealing scheduling algorithms.
- Apply machine learning to predict optimal scheduling strategies for given workload characteristics.
- Benchmark on real-world workloads such as scientific simulations, image processing, or matrix computations.

7. References

- OpenMP Official Documentation: https://www.openmp.org/specifications/
- OpenMP Examples and Best Practices: https://github.com/OpenMP/Examples

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- Task Scheduling in Parallel Systems – ScienceDirect: https://www.sciencedirect.com/
- Google Colab: https://colab.research.google.com/